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EXECUTIVE SUMMARY OF THE THESIS

Characterisation of GaAs:Cr Microstrip Sensors for Single Photon Counting

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1. Introduction

The MYTHEN III detector is a one-dimensional hybrid single photon counting (SPC) microstrip detector developed at the Paul Scherrer Institute by the Laboratory for X-ray Nanoscience and Technologies group. It was primarily designed to meet the demands of time resolved X-ray Powder Diffraction experiments at synchrotron radiation facilities [1]. The standard MYTHEN III detector employs silicon microstrip sensors. However, although the latter provides noteworthy performance, its absorption efficiency decreases at higher X-ray energies. Therefore, high Z semiconductor sensors can be used to extend the efficient detection range of the system, with chromium compensated GaAs representing one possible alternative [2] [3]. Indeed, a 500 μm thick GaAs:Cr sensor exhibits an absorption efficiency of approximately 98% at 30 keV [2], compared with approximately 15% for a silicon sensor of the same thickness. However, the use of high Z material is challenging, due to larger leakage currents, reduced material uniformity and consequent different charge transport properties [3].

This work therefore delves into the study of a MYTHEN III detector equipped with two prototype 500 μm thick GaAs:Cr sensors with 50 μm and 100 μm strip pitches. The main objective is to characterize the detector performance in terms of figures of merit like gain and noise and how the readout settings affect them, as well as how these settings influence the count rate performance at high photon rates. Moreover, charge transport properties will be investigated as well, motivated by the unexpectedly large charge sharing observed for the 100 μm strips. Eventually, the spatial resolution of the GaAs:Cr microstrips will be addressed, with particular attention to the usage of the interstrip interpolation mode.

2. Detector and experimental methods

In this work, measurements and tests have been performed using a reduced prototype of the complete MYTHEN III module [1], made up of two readout chips connected to sensors with 50 μm and 100 μm strip pitch, respectively. Each readout chain includes a charge sensitive preamplifier (CSP) followed by a shaper, as well as a set of three comparators coupled with three independent 24

bit counters [1]. In standard counting mode, a photon induced pulse is counted only when it crosses the selected comparator threshold during the exposure time. Instead, the interstrip interpolation mode makes the most out of the charge sharing phenomenon: two comparators per channel are active, with a high threshold to identify the photon event and a low threshold to detect shared charge in neighbouring strips. Depending on the coincidences, the event is assigned to the left, central or right counter, thus providing positional information below the physical strip pitch [4]. The interpolation mode could not be used with the 100 µm pitch sensor, since the readout chip was designed for 50 µm pitch, thus leaving every second channel unconnected.

The characterization was carried out using different experimental setups. A conventional X-ray tube was used as the primary radiation source, while a set of secondary fluorescence targets was employed to obtain quasi-monochromatic X-ray radiation at selected characteristic energies (thanks to the material fluorescence). In some cases, such as the high rate characterization, the direct polychromatic beam of the X-ray tube was employed in order to reach higher fluxes. Moreover, for some measurements a laser setup has been employed to generate charge carriers at controlled positions inside the sensors, allowing spatially resolved investigations of charge collection and charge sharing. Due to the opaque Ni backside, the 660 nm laser measurements were mainly performed in an edge-on configuration, restricted to a superficial lateral region.

The detector response was characterized through threshold scans, consisting in progressively varying the global comparator threshold and recording the number of counts at each step. Since SPC detectors only count pulses crossing the selected threshold, the resulting response represents an integral spectrum. For quasi-monochromatic radiation, the ideal step-like response is broadened by noise into a sigmoid-shaped curve, also called *S curve*, while charge sharing introduces an approximately linear contribution [2] [5]. For each channel, the photon induced component is fitted using a cumulative Gaussian model including charge sharing and background terms. The fit returns the inflection point (µ), the Gaussian width (σ) and the charge sharing parameter (C_s). The inflection points, which correspond to the photon energy, measured at several known photon energies, are used to build the energy calibration relation, from which gain and offset are obtained [2]. The Gaussian width, expressed in DAC units, is converted into equivalent noise charge (ENC, in electrons) using the gain and the pair creation energy (4.2 eV for GaAs [2]), while the charge sharing parameter is used to estimate the fraction of shared events. Moreover, the number of counts is extracted from the scan with threshold set to half the photon energy [4].

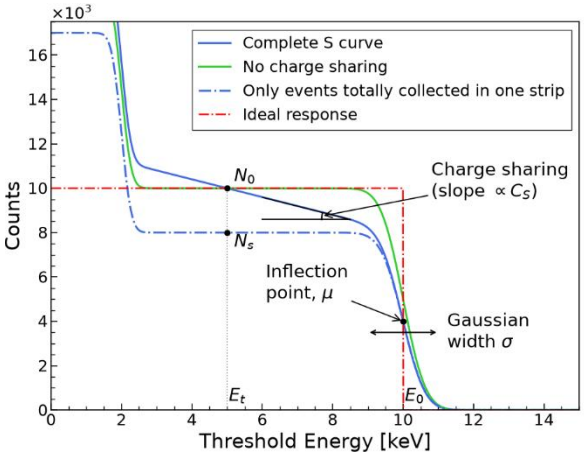


Figure 1 Example of simulated threshold scan for 10 keV photons: the abrupt increase (of the complete *S curve*) on the lower energies side is due to noise counts.

Since these parameters are extracted for several channels, a weighted mean is employed, assigning larger weights to measurements with smaller uncertainties. The uncertainty on the weighted mean describes the precision of the average detector response, while the channel to channel spread, defined as the standard deviation of the considered channels, quantifies the non uniformity of the response. In this way, useful figures of merit such as gain, noise and charge sharing have been computed.

3. Readout characterisation

The first part of this work delves into the investigation of the influence of the readout, specifically preamplifier and shaper feedback parameters, on figures of merit such as gain, noise and count rate performance, by systematically varying the different settings: indeed, several combinations of feedback capacitances were tested.

Figure 2 shows the simplified architecture of one MYTHEN III analogue chain, including the configurable feedback elements of the preamplifier and shaper stages. The equivalent feedback resistances are controlled through MOSFETs: decreasing the corresponding voltages increases their equivalent resistance.

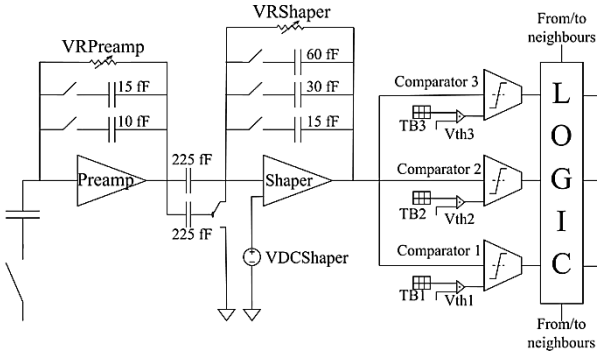


Figure 2 Simplified architecture of readout chains

For the mentioned model, assuming a negligible collection time, the ideal gain can be approximated as follows, where $C_{f,pre}$, $C_{f,sh}$ and C_{AC} represent respectively preamplifier, shaper and coupling capacitances:

$$G \propto \frac{1}{C_{f,pre}} \frac{C_{AC}}{C_{f,sh}} \qquad (1)$$

In contrast, no model has been adopted for the ENC, since it depends on several contributions, (such as shaping time, input and detector capacitance, leakage current).

According to the experimental results, a clear monotonic decrease of the gain has been observed when increasing the shaper and preamplifier capacitance and decreasing the AC coupling capacitance (the one laying between the two stages). For the 50 μm strip-pitch sensor, the gain decreased approximately from 66 to 14 DAC/keV with increasing shaper capacitance, and from 41 to 24 DAC/keV with increasing preamplifier capacitance, while doubling the AC coupling capacitance increased the gain from about 25 to 41 DAC/keV (similar results have been obtained for the 100 μm). Indeed, a comparison between the predicted (based on the Eq. ((1)) and observed relative gain reproduced the same qualitative dependence, although the variations were systematically weaker than the expected ones, consistently with the presence of non idealities. Regarding the feedback resistances, a stronger gain dependence was observed for the preamplifier one, whose increase resulted in higher gain, from approximately 22 to 50 DAC/keV for the 50 μm sensor. Conversely, varying the shaper one produced much smaller gain variations, typically around 1 DAC/keV. Although the feedback resistances do not appear in Eq. (1), they control the temporal response of the analogue chain and can indirectly affect the measured peak amplitude through ballistic deficit. The stronger influence of the preamplifier resistance is consistent with its role in the initial charge integration.

Instead, the ENC generally increased with the preamplifier feedback capacitance, from approximately 210 to 295 e^- when increasing it from 10 to 25 fF (at fixed shaper capacitance of 30 fF). Similarly, increasing the shaper one from 15 to 105 fF led to an ENC increase from approximately 190 to 275 e^- (at fixed preamplifier capacitance of 10 fF), with a tendency towards saturation at higher values. Regarding the feedback resistances, the ENC generally decreased with the increase of the latter: indeed, the strongest dependence was observed for the preamplifier resistance, with ENC decreasing from approximately 315 to 180 e^-. Conversely, increasing the shaper one produced smaller ENC reductions, typically around 10–20 e^- over the investigated range.

Overall, these measurements are qualitatively consistent with previous MYTHEN III studies performed with Si sensors [1]. Nevertheless, a direct quantitative comparison should be made with caution due to the different sensor materials, signal polarities and readout configurations. In general, however, the GaAs:Cr prototype exhibited higher ENC values than the lowest values reported for Si.

In addition, dedicated measurements performed at different temperatures revealed an unexpected dependence of the detector response. For instance, the gain increased monotonically with temperature, whereas the ENC decreased from 310.6 ± 1.9 e^- at 17 °C to 241.0 ± 2.1 e^- at 32 °C, as indicated by the FPGA temperature sensor. This behaviour differs from previous measurements with Si based MYTHEN III [1]. Possible contributions from both the sensor and the analogue readout electronics were considered, although their individual effects could not be conclusively identified.

Finally, a dedicated threshold trimming procedure was implemented: since the same global threshold is applied to all channels, local adjustments through six bit trimbits allow the effective threshold of each comparator to be finely corrected, compensating for channel to channel response variations [1]. This procedure reduced the threshold dispersion from initial values between 132.5 and 200.2 to only 4.2–5.9 equivalent electrons across both sensors and the three comparators.

4. High rate performance

The high rate behaviour has been investigated using the polychromatic emission of the X-ray tube, with the main goal of determining its dependence on the analogue readout settings.

Due to the polychromatic beam, no unique photon energy could be assigned. Therefore, the count rate was extracted from the threshold scans at a threshold set to half the effective mean photon energy, estimated from a simulation of the filtered X-ray spectrum. The counting performance was then evaluated using an effective relative efficiency, defined as the ratio between the measured count rate and the expected one, obtained by linearly fitting the low-flux response and extrapolating it to higher fluxes. For each readout configuration, the measured count rate corresponding to a 10% counting loss (due to pile up) was eventually extracted as a figure of merit. Furthermore, one channel per sensor has been considered as representative, since the direct beam did not uniformly illuminate the detector, while this was sufficient to investigate the qualitative dependence on the readout settings.

The resulting effective count rates ranged from 1.70 to 2.88 MHz for the chosen channel of 50 µm sensor, and from 1.46 to 2.60 MHz for the 100 µm one, depending on the readout configuration. These values should not be directly compared with absolute count-rate capabilities measured using monochromatic beams and independently monitored incident photon fluxes.

The clearest dependence was observed for preamplifier resistance: indeed, decreasing its value reduced the counting losses and thus the pile up, whereas increasing the preamplifier capacitance led to a greater pile up effect. This behaviour is consistent with a faster preamplifier recovery when decreasing its feedback time constant ($\tau_{pre} = R_{f,pre}C_{f,pre}$), then potentially reducing pile up. However, the effective pulse depends also on the sensor charge collection time. On the other hand, varying the shaper feedback parameters produced only minor, if not negligible, changes in the high rate behaviour.

5. Readout performance trade-offs

Overall, the combination of gain, noise and high-rate measurements showed that the preamplifier stage plays a central role in the detector response: indeed, within the investigated ranges, the decrease of preamplifier feedback capacitance increased the gain, reduced the ENC and improved the high rate behaviour. Instead, the preamplifier feedback resistance showed a clear trade-off: higher resistance values provided larger gain and lower ENC, whereas lower resistance values improved the high rate response. Regarding the shaper, only its capacitances showed effects on the counting performance. Eventually, the choice of readout settings depends on the required balance between gain, noise and count rate capability.

6. Charge transport and sharing

The second part of this work delves into the investigation of the charge sharing by analysing the *S curves* obtained with X-ray fluorescence, using the simplified linear charge collection model proposed by Bergamaschi et al. [5]. In this model, the charge collected by a strip decreases approximately linearly near its boundaries, where the generated charge cloud is shared with neighbouring channels. The charge sharing fraction therefore represents the fraction of the strip pitch occupied by this region, assuming uniformly distributed photon interactions.

For a reference 50 µm silicon sensor, a charge sharing of 39.5 ± 0.1% was measured. Significantly larger values were obtained instead for the GaAs, reaching 81.4 ± 0.1% and 74.6 ± 0.1% for the 50 µm and 100 µm pitch sensors, respectively. While the former agrees with previous results reported by Ruat et al. [2], the latter is unexpectedly high: previous measurements on 100 µm GaAs:Cr strips showed charge sharing of approximately 50% [2]. Thus, doubling the strip pitch produced a much

smaller reduction than expected from simple geometrical scaling.

To further investigate these results, spatial scans were performed using a 660 nm laser in an edge-on configuration. By progressively moving the injection position across the strips (through the sensor's lateral surface along its thickness) and extracting the *S curve* inflection points, the spatial charge collection profiles were reconstructed. Both sensors exhibited approximately trapezoidal profiles, with fitted central plateau widths of 11.08 ± 0.03 µm and 19.40 ± 0.01 µm, respectively, as visible in Figure 3.

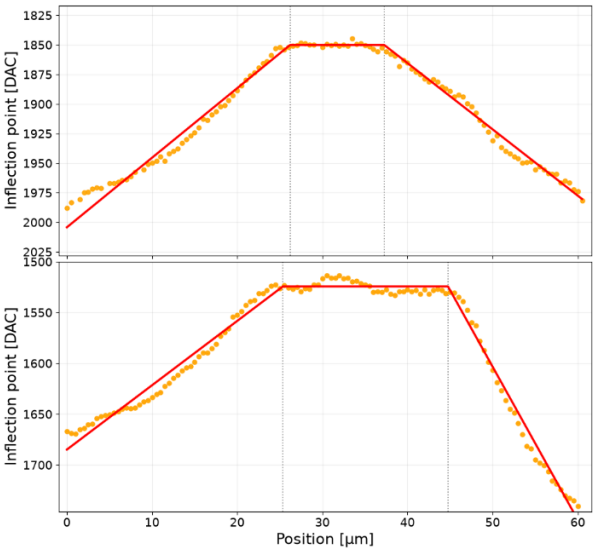


Figure 3 Spatial charge collection profiles for the 50 µm (top) and 100 µm (bottom) sensors. Orange points represent the inflection points at different laser positions, while red lines show the trapezoidal fits

The profiles shown in Figure 3 support the simplified linear charge collection model described in Bergamaschi et al. [5]. Moreover, the narrow central plateaus indicate charge sharing of approximately 80% for both sensors, in qualitative agreement with the values previously obtained from X-ray *S curves*. However, since the 660 nm laser generates charge close to the illuminated surface, unlike X-rays interacting at different depths, the comparison remains qualitative.

Eventually, the unexpected pitch dependence may be related to differences in electrode metallization and interstrip geometry compared with the sensors previously studied by Ruat et al. [2]. Such differences could affect the electric field distribution and charge collection. However, this explanation remains a hypothesis which would require further investigation.

Finally, the crater effect was qualitatively investigated by focusing the 660 nm laser at different positions along the sensor thickness. The crater effect consists of a reduced signal in the collecting strip, alongside an induced one of opposite polarity in the neighbour strips, due to incomplete signal induction caused by the short hole lifetime in GaAs, as described by Greiffenberg et al. [3]. Evidence of this phenomenon was observed in the performed threshold scans, where *S curve*-like responses appeared on the opposite side of the electronic baseline when injecting charge close to the readout electrodes. These results are qualitatively consistent with the crater effect, although its magnitude was not quantified.

7. Spatial resolution

The final part of this work covers the characterisation of the spatial response of the GaAs:Cr sensors by using laser scans to obtain the Point Spread Function (PSF), which describes the probability of detecting a photon as a function of its interaction position. The corresponding Modulation Transfer Function (MTF), describing the detector's ability to preserve contrast at different spatial frequencies, was calculated from the normalized magnitude of the PSF Fourier transform [4].

In standard counting mode (STD mode in Figure 4), the first MTF minima were found at 19.98 lp/mm for a representative channel of the 50 µm-pitch sensor, and 8.81 lp/mm for the 100 µm one: these values agree with the ones expected from their physical pitches.

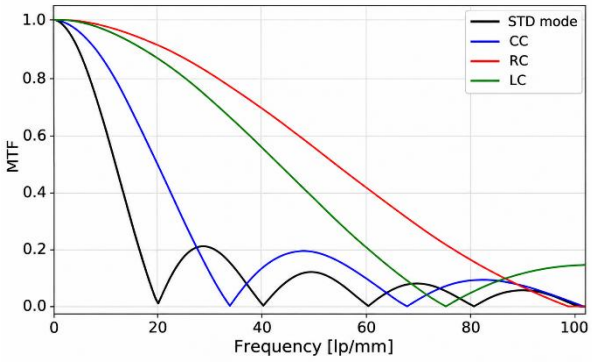


Figure 4 Comparison of the MTFs obtained in standard and interstrip interpolation modes for a representative channel of the 50 µm-pitch sensor

Instead, in interstrip interpolation mode (available only for the 50 μm pitch sensor in this prototype), charge sharing is exploited to assign photon events to left, central and right counters (Bergamaschi et al., [4]). The resulting PSFs were narrower than in standard mode, with the central counter (CC) exhibiting a first MTF minimum of 33.85 lp/mm, compared with 19.98 lp/mm in standard mode for the selected channel. The side counters (RC and LC) exhibit even higher first-minimum frequencies due to their narrower spatial responses, with some asymmetry possibly arising from residual differences in threshold response of comparators, even after trimming.

8. Conclusions

Overall, this work demonstrated the feasibility of operating GaAs:Cr microstrip sensors with the MYTHEN III single photon counting detector, providing a promising solution for extending its efficient detection range towards higher X-ray energies. The measurements highlighted the flexibility of the analogue readout, with the preamplifier settings playing a central role in determining gain, ENC and high rate behaviour. No single configuration was identified as universally optimal, since the settings should be adjusted according to the required balance between gain, noise and count rate capability. Furthermore, the successful threshold trimming and the measured effective count rates support the practical potential of this configuration.

One of the most significant findings was the unexpectedly large charge sharing observed in the 100 μm pitch sensor, only moderately lower than that of the 50 μm one. The laser measurements supported these results and provided qualitative evidence consistent with the crater effect. Nevertheless, the origin of the unusually large charge sharing remains to be fully understood, with differences in electrode geometry representing a possible explanation.

The spatial response measurements showed that the substantial charge sharing observed in GaAs:Cr, although potentially detrimental in standard counting mode, can be exploited by MYTHEN III interpolation to achieve a narrower effective spatial response and improved position discrimination.

In conclusion, the results indicate that GaAs:Cr sensors can extend the potential of MYTHEN III towards higher photon energies while preserving the main advantages of single photon counting, including the possibility of improving spatial resolution through on-chip interpolation.

References

[1] M. Andrä et al., "MYTHEN III: advancements in single photon counting detectors for synchrotron powder diffraction experiments," *Journal of Synchrotron Radiation,* vol. 32, pp. 365-377, 2025.

[2] M. Ruat et al., "Photon counting microstrip X-ray detectors with GaAs sensors," *Journal of Instrumentation,* vol. 13, no. C01046, 2018.

[3] D. Greiffenberg et al., "Characterization of Chromium Compensated GaAs Sensors with the Charge-Integrating JUNGFRAU Readout Chip by Means of a Highly Collimated Pencil Beam," *Sensors,* vol. 21, no. 1550, 2021.

[4] A. Bergamaschi et al., "First demonstration of on-chip interpolation using a single photon counting microstrip detector," *Journal of Instrumentation,* vol. 17, no. C11012, 2022.

[5] A. Bergamaschi et al., "Looking at single photons using hybrid detectors," *Journal of Instrumentation,* vol. 10, no. C01033, 2015.

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